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Cellular-scale hydrodynamics

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Abstract
Microfluidic tools are providing many new insights into the chemical, physical and physicochemical responses of cells. Both suspension-level and single-cell measurements have been studied. We review our studies of these kinds of problems for red blood cells with particular focus on the shapes of individual cells in confined geometries, the development and use of a 'differential manometer' for evaluating the mechanical response of individual cells or other objects flowing in confined geometries, and the cross-streamline drift of cells that pass through a constriction. In particular, we show how fluid mechanical effects on suspended cells can be studied systematically in small devices, and how these features can be exploited to develop methods for characterizing physicochemical responses and possibly for the diagnosis of cellular-scale changes to environmental factors.

(Some figures in this article are in colour only in the electronic version)

1. Introduction

1.1. Objectives

The goal of this paper is to present a short review of our research that reports microfluidics-based experiments and modeling for flows of suspension of red blood cells (RBCs) and fluid dynamics measurements at the scale of single cells flowing in confined spaces. The subject is of active interest in many laboratories around the world and there are many different approaches being taken and systems being studied. In particular, the microfluidic toolbox offers routes for basic research investigations of cellular-scale physics and chemistry, for cell sorting and various combinatorial strategies, for designing tools for sensing and measurement even at subcellular levels, and for drug delivery. The individual devices span from a few to hundreds of microns (the associated circuitry and hardware inevitably make the final package larger), and the microfluidic tools are used to help address questions from the scale of a single cell to the tissue and organ scale if not beyond.

The widespread use of microfluidic methods for probing basic chemical and physical questions of cellular systems, from the collective effects of many cells to the response of individual cells, even at the sub-cellular level, has transformed the research in these fields. Moreover, there have been many innovative ideas for using these approaches to ask new kinds of research questions and to consider possible ways to impact new technologies, e.g. see one or more of available reviews of the basic ideas, fluid dynamics and transport processes in these systems (e.g., Whitesides and Stroock (2001), Stone *et al* (2004), Squires and Quake (2005)). Although some methods utilize common laboratory materials (glass, silicone), most if not all of the approaches are amenable to development with biocompatible materials. In addition, the ability to visualize individual cells under flow conditions using high-speed imaging coupled with modern image-processing tools offers new insights particularly for thinking about individual cells in confined geometries. For example, these kinds of

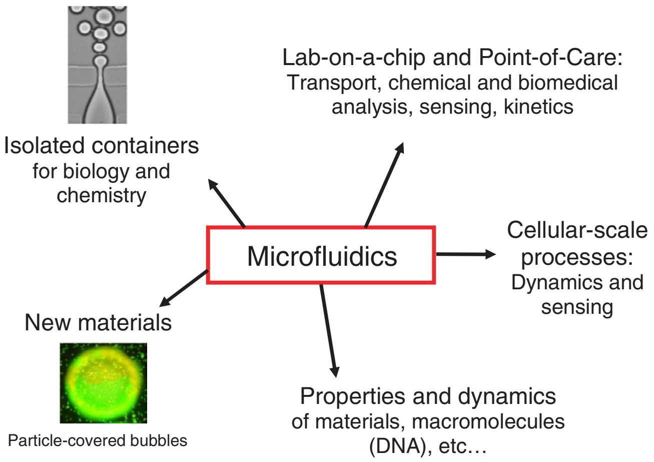

Figure 1. One view of the landscape for the use of microfluidic tools for the study and control of biologically relevant materials and systems. Different areas of research are indicated, including those focused on fundamental science questions as well as those directed toward technologically relevant materials and devices. The image in the upper left shows the individual formation of liquid drops in a flow-focusing device (Anna *et al* 2003), which is one example of the controllable formation of monodisperse micrometer objects using microfluidic methods. The image in the lower left shows a particle-covered gas bubble that was made in a microfluidic device (Subramaniam *et al* 2005).

methods have been used to develop mimics of the chemical dynamics of blood clotting (Kastrup *et al* 2006), address questions of developmental biology (e.g. Lucchetta *et al* 2005), demonstrate subcellular delivery of chemicals (Takayama *et al* 2001), probe the role of mechanical stresses from fluid shear in producing cellular-level injury (e.g., Huh *et al* 2007), control the distribution of chemicals with a biocompatible gel (Choi *et al* 2007), investigate the occlusion of vessels that occurs with sickle cells (Higgins *et al* 2007), etc. Finally, the developments include integrating tools for manipulating fluids with the hardware available for modern integrated circuits to make new kinds of devices with unique functionality (Lee *et al* 2007).

One more general view of microfluidic approaches for the study of biologically and medically relevant questions, admittedly from the perspective of material science, engineering and physics, is given in figure 1. Thus, it is possible to use these systems to handle cells in new ways (e.g., put them in individualized containers such as liquid drops (e.g., Thorsen *et al* (2001), Anna *et al* (2003)), make new layered types of materials such as particle-covered bubbles (Subramaniam *et al* 2005), or develop micron-scale materials for identification and diagnosis such as is possible with bar-coded particles suspended in solution (Pregibon *et al* 2007). Given the kinds of independent measurements that are evident in the themes represented in figure 1 it is natural to think about these from the view of 'point-of-care' (POC) diagnostics, which stand at the intersection of biomedical analyses and microfluidic technologies. For example, doctors rely on blood tests to assess a patient's health and to diagnose and monitor diseases. Some tests measure the components of blood itself; others examine substances found in the blood in order to identify possible abnormal functioning of an organ. For those POC devices dedicated to biomedical analyses on blood samples, i.e. blood counts, dosage of plasma proteins (e.g., transaminase, cholesterol, glucose, etc), one step is usually the separation of plasma from blood cells, and both this separation as well as measurements on individual cells can be accomplished using microfluidic devices. In addition, significant advances have been made in the design of artificial blood, e.g. submicron dimension lipid vesicles containing hemoglobin (e.g., Chang (1999)) and systems have been tested on animals.

1.2. Hydrodynamics of microcirculation

Given the focus of this paper on small devices for handling fluids and, in particular, suspensions of cells, we thought it useful to provide a very brief introduction to some of the important physical ideas needed to describe and characterize these flows. Hydrodynamics of blood in the microcirculation is a laminar bounded viscous flow. A basic idea in characterizing fluid motions is to report the Reynolds number, $Re = UH/\nu$, where ν is the kinematic viscosity of the fluid (i.e., the ratio of the shear viscosity η and the density ρ of the fluid), U is the typical linear speed of the flow, and H is the typical distance over which the gradients of velocity are the highest (usually the diameter of a channel for a confined flow or the height of the channel). Physically, the Reynolds number measures the relative importance of inertial to viscous forces. In natural microvessels or in the laboratory microchannels we used in our studies, $Re < 1$, e.g. we typically have $0.001 < Re < 0.1$. Hence viscous (frictional) effects are most significant in thinking about the physics of these small-scale flows.

The driving force for blood flow in the body is the pressure produced by the pumping of the heart. In the microvessels the flow results from a competition between this peristaltic pressure and the viscous friction on the walls of the blood vessels. Because the Reynolds number is small for the microcirculation, the flow is laminar, and the pressure in a steady flow decreases linearly with distance; the resulting pressure drop $\Delta P = RQ$, where Q is the volumetric flow rate and the coefficient of proportionality R is called the hydrodynamic resistance of the microvessel. In the case of a rectangular channel common to microfluidic approaches, the hydrodynamic resistance is given approximately by $R = 12\eta L/WH^3$ (for $H \ll L$), where L is the length of the channel, W is the width and H is the height.

The linear relationship between ΔP and Q is reminiscent of Ohm's law linking the difference of electric potential to the current passing through a resistor of resistance R (for a brief introduction to microfluidics written for electrical engineers and others unfamiliar with fluid systems, see Stone (2007)). When thinking about the average flows in networks that are common to microfluidic arrangements, it is often convenient to think about the analogous electrical network, e.g. hydrodynamic resistances in parallel or series can be added similarly to electric resistances in parallel or series. For example, when two channels of different sizes follow each other the global resistance of the system is the sum of each channel's resistance and the effect on the flow is dominated by the channel with the largest hydrodynamic resistance. Moreover, when a small object such as bead, drop, bubble or cell enters a channel the resistance along the channel is given by the sum of the channel's resistance in the absence of the particle and the resistance developed across the length of the object, which depends on the local characteristics of the flow and the viscoelastic properties of the object (e.g., see figure 8).

Finally, it is common to characterize the influence of the flow on the walls of blood vessels, channels or suspended cells with the local fluid shear rate or the shear stress the fluid exerts on a surface. In several of the studies we describe below the shear rate or shear stress are convenient measures of the ability of the flow to locally deform objects. Using the scales introduced above the shear rate is simply estimated as U/H and the shear stress is estimated as $\eta U/H$, where H is the smallest dimension of the flow system (since it is across this distance that the velocity gradients are largest).

1.3. Outline of the paper

Our focus has been on questions where hydrodynamics, deformability and elasticity may play some role and we give several examples in this paper. In particular, in section 2 we summarize our experiments on the shapes of individual RBCs in confined spaces, including various microfabricated geometries that can cause substantial and repeated deformations of the cells. In this first set of studies our focus is more on qualitative features though where possible we reference related quantitative ideas. In section 3, to understand better the deformation sequence of an individual cell, we used the versatility of microfluidics to propose a simple method, which we call a differential manometer, to measure the variations of the pressure drop induced by the passage and the deformation of cells in a narrow channel. We illustrate the idea for looking at individual cell lysis events under flow. Finally, in section 4 we consider a problem at the scale of a suspension of cells. In particular, we illustrate how a constriction in the flow geometry produces cross-streamline migration of cells. The consequence of this dynamical effect is the significant increase of a cell-free layer downstream of such a constriction. A technological application of this effect to the separation of cells from plasma is illustrated. Throughout the paper we indicate further avenues for studies of dynamics and hemorheology.

2. Flow of RBCs in single capillaries

There is a long history of the study of cells at the micrometer scale, and such approaches have been used to assess the mechanical properties of cells (Mohandas and Evans 1994), for observing cell shapes in the microcirculation (Gaehtgens *et al* 1980), and for improved understanding of microcirculatory diseases (e.g., Chien *et al* (1971), Nash (1990)). For example, studies that characterize the shapes of individual cells in confined geometries may be useful for indicating changes in the surface area of the cell in contact with the capillary walls, which can impact blood–oxygen exchange and/or adhesive properties, and can be altered by chemical–mechanical changes of individual cells such as that often occur in disease. In this case, the main features during flow are parachute-like cell shapes, which have been understood physically from the interplay of bending and stretching forces of the cell membrane and the lubrication forces of the fluid passing in the narrow gap between the cell membrane and the capillary walls (e.g., Secomb *et al* (1986)).

Moreover in vivo and in vitro observations (Bagge *et al* 1980, Gaehtgens *et al* 1980) demonstrated the existence of non-axisymmetric 'slipper-like' shapes. One analytical study (Secomb and Skalak 1982) showed how prescribed two-dimensional asymmetric slipper shapes require a smaller pressure drop for a given cell velocity than symmetric 'parachute-like' shapes as a consequence of the slow tank-treading of the membrane. Recent advances in computer simulations calculated the evolution of either quasi-spherical (Pozrikidis 2004) or biconcave elastic capsules (Noguchi and Gompper 2005) in capillary flows. In low velocity regimes (speeds mm s^{-1}), the main results were that the transition from a discocyte shape to a slipper-like or parachute-like shape depended both on the velocity (Noguchi and Gompper 2005) and the position of the center of mass of the cell relative to the centerline of the flow (Pozrikidis 2004). Basically, in low velocity regimes the bending resistance of the membrane is important compared to shear stresses and a slipper-like shape can be observed for velocities that are very low. Nevertheless, for a sufficiently high velocity regime (speeds much greater than mm s^{-1}) and for small enough capillary radii (less than 5 μm), such as those obtained in filtration experiments for example, no numerical studies exist since local curvatures at the rear of the cell can become very high and numerically intractable; in this limit only axisymmetric approaches have been attempted (Secomb *et al* 1986). With respect to this higher speed regime, the behavior of cells in confined environments and at high shear stresses is also of importance in some aspects of hemolytic anemias associated with a reduction of the lumen of small arterioles (microangiopathic anemias, e.g. Chien *et al* (1971)). Given the dearth of studies, in one set of experimental studies we sought to better characterize single-cell flow behavior in this high velocity regime. Here we report results obtained using high-speed visualization of flow in micrometer diameter glass capillaries. We also consider flows in other micrometer flow configurations.

Microfluidic approaches coupled to high-speed video imaging processing allow the whole sequence of the deformation of a cell entering a microchannel of smaller size to be measured with great detail; it may even offer a way to characterize the influence of possible diseases on RBCs. Other approaches for characterizing a population of cells exist such as the filtration technique (Nash 1990, Skalak *et al* 1983, Fischer *et al* 1992, Drochon 2005), which consists in flowing the cells through a multipore polycarbonate membrane: the mean passage time of the cells or the measured pressure-drop versus flow-rate relation provides an indication of the average state of deformability in a population of cells. Nevertheless, the traditional techniques are not able to resolve information at the scale of a particular pore or a single cell.

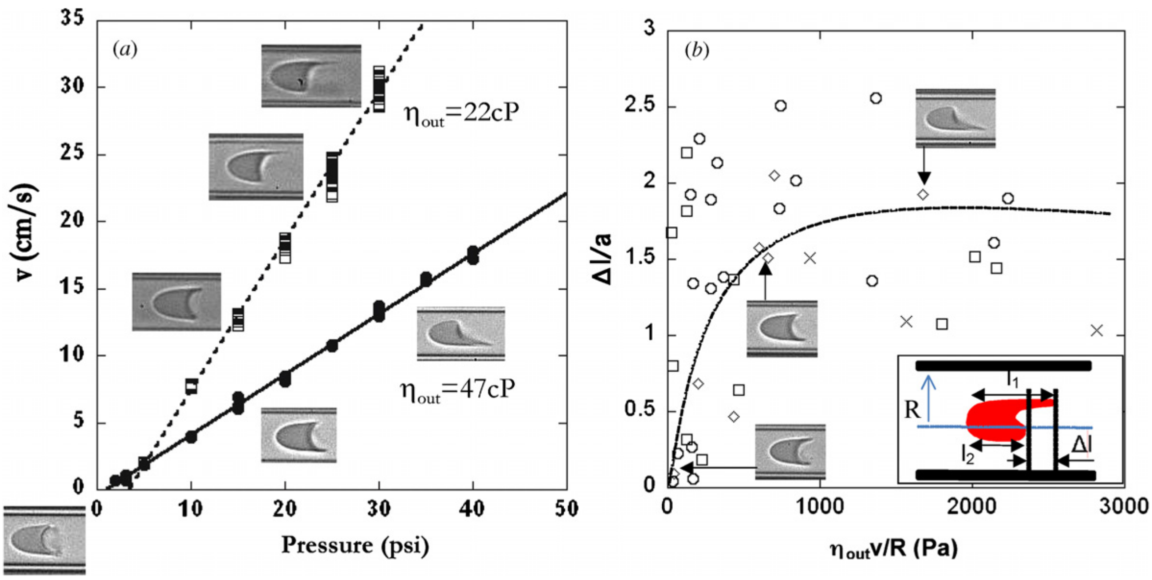

Figure 2. (*a*) RBC velocity v as a function of the injection pressure applied at one end of the capillary tube. Here $R = 8.9$ μm. The minimum applied pressure necessary for the cells to move is due to the relative height of the injection syringe compared to the device. (*b*) The ratio of the difference of length Δl (see inset) between the two sides of the slipper and the size a of the red blood cells as a function of the shear stress $\eta_{out}v/R$: (○) $0.3 < a/R < 0.4$, (□) $0.4 < a/R < 0.6$, (◊) $0.7 < a/R < 0.8$ and (×) $0.8 < a/R < 1$. Images of three cells are also displayed and have the same equivalent radius $a = 3.1$ μm; they flow in capillaries of the same radius $R = 4.2$ μm. The dashed line is a guide for the eyes.

2.1. Circular glass capillaries: from parachute-like to slipper-like shape

In this section, we establish a picture of the shapes of RBCs flowing in small capillaries as a function of both flow and geometric parameters. The different parameters varied during our experiments are the mean speed v of the red blood cells, the viscosity η_{out} of the suspending medium and the ratio between cell's size and the capillary radius R. The cell size a is estimated from the surface area of the disc of the same projected area S of the flowing cell: $a = (S/\pi)^{1/2}$. The capillaries were several centimeters in length and the observations were made 0.5–1 cm from the exit. All the results reported below are stable shapes. Images illustrating the shape evolution are presented in figure 2(*a*).

2.1.1. Effect of cell velocity and external viscosity. We investigated the effect of η_{out} and v on the shape of the RBCs for capillaries with different radii R. The results are presented for the same capillary radius in figure 2(*a*) but for two different buffer viscosities $\eta_{out} = 22$ and 47 mPa s^{-1}. The results show that by increasing v or/and η_{out}, the cells assume a slipper-like shape whose 'tail' becomes longer and pointed. This result suggests that an important parameter of the transition is the mean shear stress acting on the cells. For increasing stresses, the cells develop a longer tail. We estimated the relative elongation of the tails compared to the size a of the cells as a function of $\eta_{out}v/R$ as represented in figure 2(*b*). The deformation of the cells seems to saturate for stresses of the order of 500–1000 Pa.

2.1.2. Transition from centered to non-centered position. The transverse position and the shape of the red blood cells depend on their confinement during flow, characterized by the ratio a/R. Hence, for given shear stresses, the cells have a tendency to be off center relative to center line of the capillary tube. In figure 3(*a*), we report the position of the center of mass of the cells as a function of a/R for different shear stress ranges. We can see clearly a transition from centered to non-centered cells for relatively low confinement for values of a/R of the order of 0.4. These results, used together with the previous observations, allow us to draw a general semi-quantitative shape diagram as represented in figure 3(*b*). Two regions appear distinctly on the diagram: a region where the shapes of the cells are parachute-like, and a second region of cells with non-axisymmetric slipper-like shapes. The transition between these two regimes is identified using the results in figure 2(*b*). The non-axisymmetric region can also be divided into two zones: either all of the cells have their center of mass lying on the center line of the flow or the cells are slightly offset. We observe in the diagram that the tails of the slippers can be as long as the length of the cell's body; nevertheless, in this regime we did not observe any irreversible deformation of the membrane.

2.2. Sequence of deformation of RBCs entering and exiting a microchannel

2.2.1. Single-cell characteristics in confined flows. We next describe some of our research focused on hydrodynamic signatures of individual cells flowing in close-fitting geometries in the form of long constrictions. First, in figure 4(*a*) we provide time-lapse images of a healthy RBC translating into and then out of a microfluidic constriction whose smallest cross-sectional dimensions are 5 μm × 5 μm. We observe that the cell is elongated as it enters

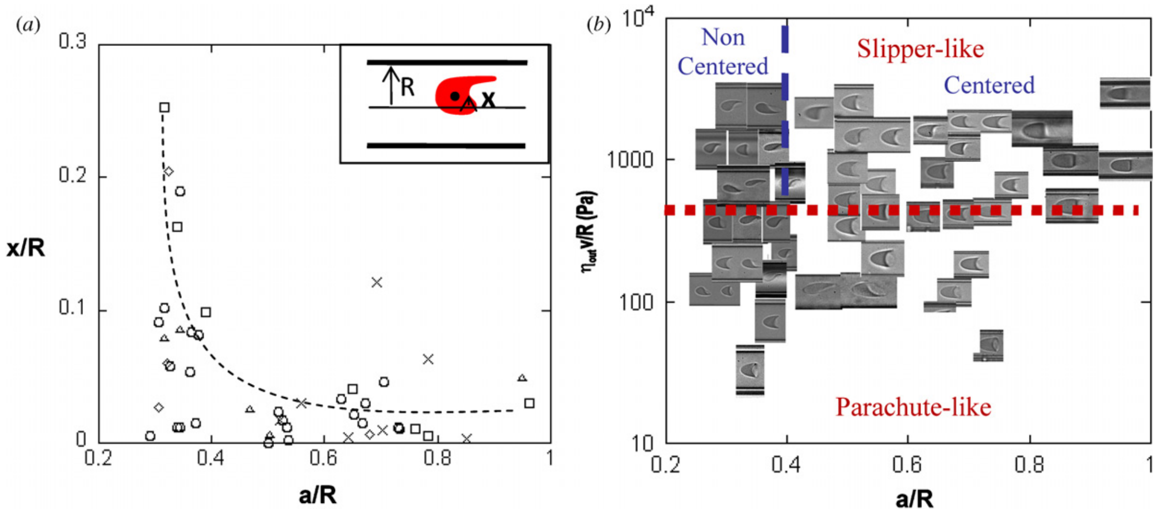

Figure 3. (*a*) Transverse position x of the center of mass of the cell normalized by R as a function of the confinement a/R (see inset for the definition of x): (○) $10 < \eta_{out}v/R < 500$, (□) $500<\eta_{out}v/R < 1000$, (◊) $1000 < \eta_{out}v/R < 1500$, (×) $1500 < \eta_{out}v/R < 2000$, and (Δ) $2000<\eta_{out}v/R$. (*b*) Phase diagram of RBC shapes as a function of the external shear stress $\eta_{out}v/R$ and the non-dimensional particle size a/R, where R is the radius of the capillary tube that is systematically changed in the experiments.

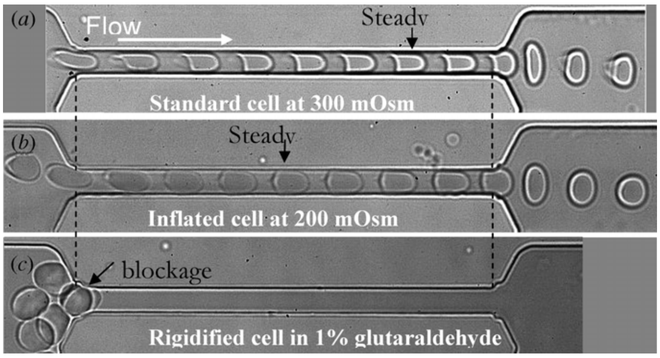

Figure 4. Time-lapse sequence of deformation of (*a*) a healthy RBC, (*b*) an inflated RBC and (*c*) rigidified RBCs flowing in a close-fitting 100 μm long channel (time lapse of 10 ms).

the channel; in the entrance region the in-plane flow is approximately a uniaxial extensional motion that succeeds in stretching the cell along the flow direction. The cell translates along the narrow channel and eventually adopts the steady characteristic 'parachute' shape (see the previous section), which it maintains until the exit region of the constriction. Near the exit the cell is again significantly stretched, though since the in-plane flow is now nearly a biaxial extensional motion, the cell is deformed perpendicular to the mean flow direction. The characteristic distance from the entrance for which the RBC reaches a stationary shape depends on its volume. Indeed, more inflated cells such as the one represented in figure 4(*b*) need a shorter channel length to reach a steady shape compared to normal cells. Moreover this transition time also depends on the mechanical properties of the cells, with the limit being a rigid cell obtained by treating the suspension with glutaraldehyde. As represented in figure 4(*c*), in the extreme case of a large nearly rigid surface, the cells can no longer enter the capillary and pack at its entrance, eventually clogging the channel.

2.2.2. Formation of trains of cells in 1D and 2D channels. One basic fact observed in the pioneering work of Gaehtgens *et al* (1980) is the formation of trains of RBCs in one-dimensional channel flow such as we present in figure 5(*a*). These authors explained that the files (or chains) of cells are formed because of the slowest cells traveling down the channels, which slow down the faster moving cells; this effect could explain why intermittent flow of RBCs is observed *in vivo* in the smallest capillaries (Johnson and Wayland 1967).

Because of the versatility of microfluidics, we can easily reproduce such a behavior (figure 5(*a*)) but more importantly we can expand and explore other flow configurations. For example, we analyzed the flow in two-dimensional channels for which the width of the channel is much larger than its height (figure 5(*b*)). Even though the lateral constraint on the movement of the cells is released, we still observe the spontaneous formation of trains of cells along the flow direction. This type of alignment cannot be easily understood by the previous argument. Rather, we believe that the specific shape taken by the cells (see figure 5(*c*)) illustrates the important friction existing between the cell and the top and bottom walls, which lead to the cells having a characteristic 'banana' body oriented orthogonally to the direction of the flow and two thin extensions of the cell's membrane extending behind due to the lubrication forces pulling the membrane (see figure 5(*c*)). This wall friction produces a relative displacement between the surrounding fluid and the cell and, in addition, for the narrow gap between two wide plates produces long-range hydrodynamic interactions between the cells.

2.2.3. Toward more complicated geometries. As a last illustration of the types of experimental studies that are possible to achieve with microfluidics, we present in figure 6 two types of flows impossible to realize with simple

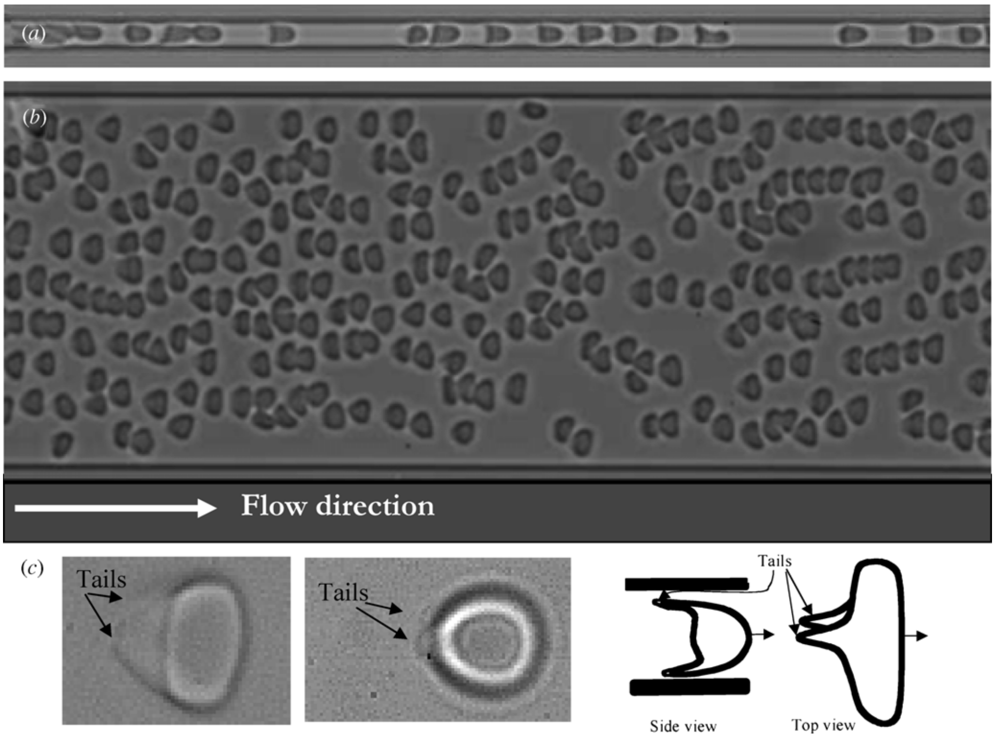

Figure 5. File of RBCs flowing in (*a*) a one-dimensional channel (100 μm long, 5 μm width and 5 μm height, at 100 μL h^{-1}) and (*b*) a two-dimensional channel (1000 μm long, 100 μm width, 5 μm height, at 100 μL h^{-1}). (*c*) In higher magnification (100×), the shape resembles a banana with two tails emanating from the back of the cell. On the left is represented a healthy cell and in the middle image an inflated cell (200 mOsm). The right image shows a schematic of the shape in side and top views.

circular glass capillaries. In figure 6(*a*) we show a 'shark teeth' type of channel—there are rapid, spatially periodic variations of the cross section—which we have used to study rapid deformations of cells. Figure 6(*a*) illustrates one cell at three different instants of time progressing from one pore to another pore; the RBCs are submitted to successive sequence of large stretching and compression whose spatial period is fixed but whose time period depends on the flow rate that is fixed independently. As demonstrated in figure 6(*a*), shark teeth geometries allow a strong and regular deformation of the cells in time. Indeed, the transit time of the cell through a contriction depends strongly on the mechanical properties of the membrane. Hence, we hypothesize that the more the cell is submitted to transient deformations, the longer would be the time needed to cross an entire field of shark teeth and so the more sensitive will be the total transit time to the mechanical properties of the membrane.

In figure 6(*b*) we show how other flow topologies can also be created with microfabrication methods; a high density of cells is shown migrating through a two-dimensional channel. In particular, the channel now contains an array of posts that separate the stream lines and can present complex dynamics of flow for RBCs. This type of arrangement may have analogies with some physiological situations but more importantly it may be relevant to different ways of handling cells in the laboratory. We note that there are several innovative developments using microfluidic channels with post arrays to effectively separate particles (e.g., Huang *et al* (2004)).

3. Differential manometer

In the previous section we illustrated different ways that high-speed video imaging and the choice of the channel geometry could give insights into behavior of individual cells in flow configurations representative of the microcirculation. In these situations it is important to ask about mechanical measurements that characterize the motion of the cells. However, rapid variations of pressure are very difficult to measure at the micrometer scale and below. In addition to issues associated with interfacing and implementing microelectromechanical system devices to standard pressure gauges (Hosokawa *et al* 2002, van der Heyden *et al* 2003, Blom *et al* 2005, Kohl *et al* 2005), the principal problem is that relatively large volumes are needed for operation of commercially available sensors compared to the volume displaced by flowing RBCs. Indeed, when a single RBC enters a channel of 5 μm × 5 μm, the volume variation produced by a flow at physiological speeds of a few millimeters per second is ~100 fL in a few milliseconds, which represents a typical pressure-drop variation of tens to hundreds of Pa.

3.1. The idea to measure changes in pressure drop

To measure simultaneously the dynamical deformation of the cells and the variation of the pressure drop produced by their motion in the channel, we developed a device with twin channels (Abkarian *et al* 2006): a test channel and an identical control, or 'comparator', channel (figure 7(*a*)), both of which produce downstream two parallel and adjacent streams of fluid;

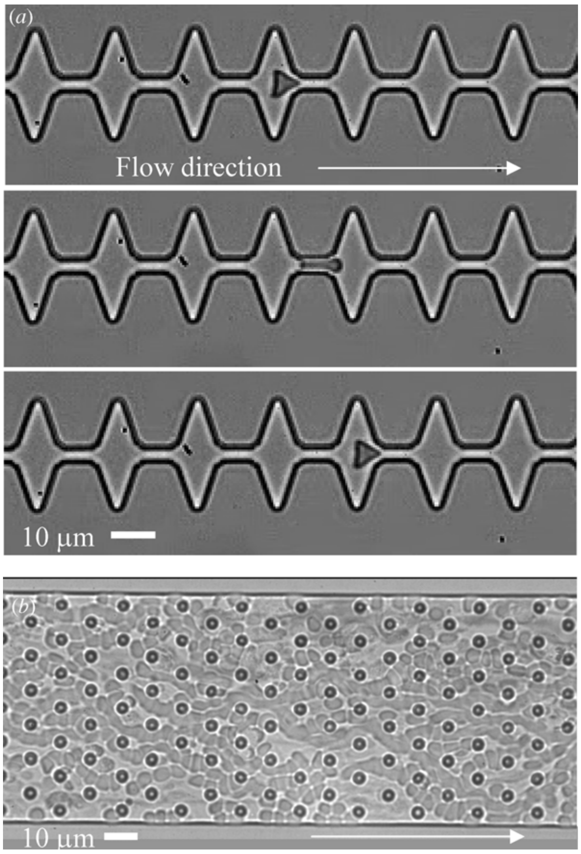

Figure 6. (*a*) Time-lapse sequence of the flow of a RBC in a shark teeth channel with 5 μm spacing between the teeth (time lapse 10 ms). (*b*) Two-dimensional channel (height $\ll$ width) with hexagonally distributed circular posts. The height of the channel is equal to 5 μm.

the idea was inspired by a similar configuration introduced by Groisman *et al* (2003). To maintain a stable interface, the two fluids are miscible, and the liquid flowing through the control channel is dyed to visualize the downstream interface (figure 7(*a*)). At steady state the downstream interface is flat but any obstruction in the test channels alters the flow rate in that channel and thus displaces the interface, whose mean position can be measured using image processing techniques (figure 7(*b*)) (all the pressure-drop values presented in Abkarian *et al* (2006) have to be divided by a corrective factor 2 because of an error in the procedure of image analysis). Hence, the principle of the measurement is based on the idea that for a given applied pressure difference across the device, the displacement of an object into and along the channel decreases the flow rate, which alters the position of the interface downstream; there is an additional pressure drop that occurs across the particle in the channel (figure 7(*c*)). The measurement of the interface deflection allows the change in pressure associated with the moving object to be determined after a basic calibration procedure without cells. Consequently, we are able to monitor the time-dependent dynamical changes in the pressure drop in the test channel.

Because the flows are laminar it is relatively straightforward to supplement the experimental studies with detailed numerical (and sometimes analytical) studies. In this case we have used numerical simulations to better understand the downstream deflection of the interface (figure 8(*a*)). In particular, the calculations verify that the deflection is linear in the change of pressure (figure 8(*b*)) at least for small pressure changes. A recent study has shown how the interface displacement in these microfluidic comparators is linked to the cross-sectional geometry of the device (Vanapalli *et al* 2007).

3.2. Measurements on different numbers of cells with variable mechanical properties

We next present how this comparator device is able to measure the change in the hydrodynamic resistance of the channel when the mechanical properties of the cells and their number density are modified. In figure 9, we compare the response of a single healthy cell with glutaraldehyde-treated cells. Glutaraldehyde is known to cross link elements that it contacts. We hence believe that it polymerizes the surface of RBC making them stiffer (e.g., Fischer *et al* (1992)). When flowed through our comparator device we observe that the excess pressure drop associated with the passage of a rigidified cell is both higher in intensity and broader in time, which reveals a slower deformation for the modified cell relative to a healthy cell for the same flow environment.

The device is also sensitive to the number of cells present in the channel. We report in figure 9 the excess pressure drop associated with the flow of 1, 2 and 5 cells (cells are closely spaced similar to rouleaux). The pressure drop systematically increases as the number of cells increases but the results are close to but not exactly proportional to the number of cells. This qualitative response is typical of confined geometries with suspended particles spaced closer than the microchannel width.

3.3. Lysis of RBCs

Using this approach we can make measurements of the membrane lysis of individual RBCs. The rupture of the membrane can then be correlated with other simultaneous measurements. For example, in figure 10 we show high-speed images of an individual RBC that has been swelled by changing the osmotic conditions as it flows through a narrow microchannel. A healthy RBC was found to always translate through the channel without damage. On the other hand, we observe that when a swollen cell enters the channel, there is a significant plugging effect, as demonstrated by following tracer particles that show that the flow through the channel abruptly decreases (see figure 10(*b*)). The cell then deforms into a prolate ellipsoidal shape while the pressure drop (reported in figure 10(*c*)) reaches a maximum. At this moment, a pore is observed to open in front of the cell; we expect that this location is where the tensile stresses are the highest (e.g., Queguiner and Barthes-Biesel (1997)). The hemoglobin jets out, allowing the flow in the channel to restart (figure 10(*b*)), and the cell continuously deflates and takes on the parachute-like shape described previously, until it becomes a transparent 'ghost'. Using our differential manometer, as the cell enters the channel we recorded simultaneously an increase

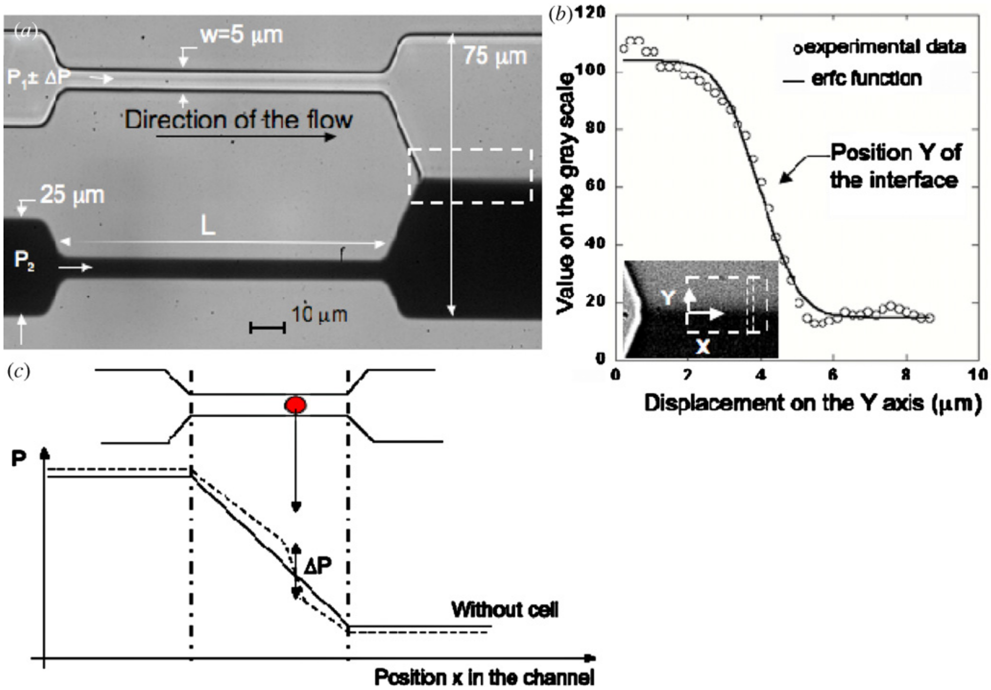

Figure 7. A differential manometer. (*a*) The specific geometry of the device allows the measurement of the additional pressure drop induced by the flow of a close-fitting cell in the upper channel. (*b*) Inset: close-up of the interface whose deflection serves to calibrate the change in pressure. Image analysis determines the variation ΔY of the position of the co-flowing line that marks the interface. (*c*) The solid line shows the pressure profile of the upper channel, in the absence of any object, where on the left is the applied pressure P_1 and on the far right is the outer atmospheric pressure. The schematic indicates that the change in the pressure due to the hydrodynamic resistance of the flowing object mainly occurs in the narrower section of the device at the particle's current location.

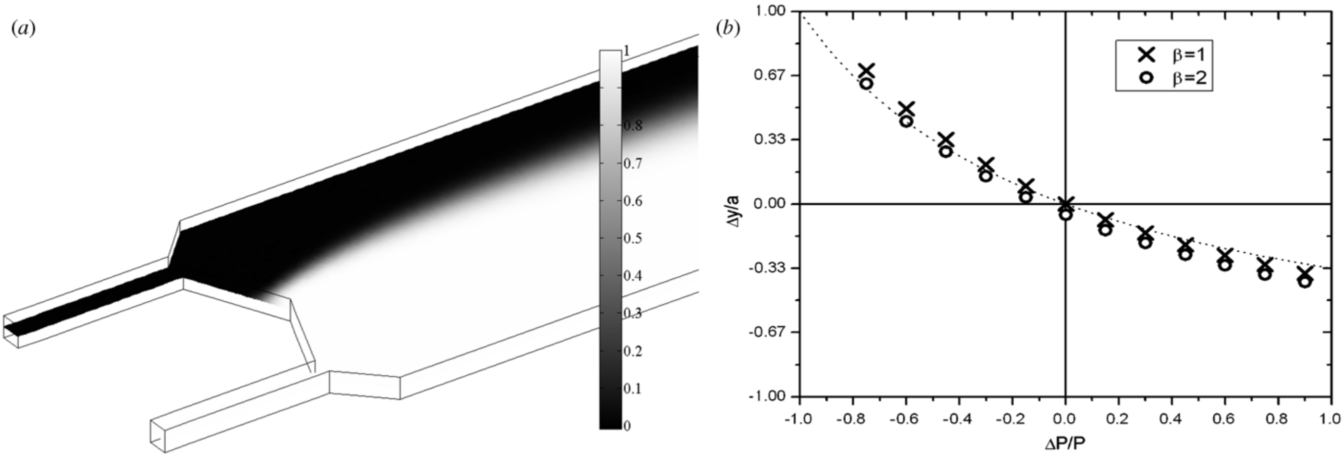

Figure 8. (a) The graphic shows the concentration field (representing dye on one side of the two-channel flow) for $\beta = \eta_{\mathrm{ink}}/\eta_{\mathrm{cells}}$ and $\Delta P/P_1 = -0{,}6$. A steady co-flow is reached 50 μm into the channel. The color gradient indicates the diffusion between the ink and the other solution. (*b*) Non-dimensional deflection Δy versus pressure difference $\Delta P/P_1$ plotted for different values of the ratio between the viscosities β. The points are Comsol Multiphysics calculations (crosses and circles), along with an approximate analytical result we developed (dotted and dashed lines).

of the pressure drop in the test channel until lysis occurs. The value of the critical stress is measured to be around 0.4 psi (this value corrects a conversion error we found in our first report of this result, Abkarian *et al* (2006)) and this value is in good agreement with approximate values obtained with static micropipette experiments on preswollen RBCs (Mohandas and Evans 1994).

These results suggest an easy way to make rapid and successive measurements of lysis and the associated change of pressure for a large number of RBCs, which would allow for statistical characterizations of this medically important phenomenon. Indeed, while micropipette techniques take several minutes at least to get one value per cell, our method allows measurement for every 10–100 ms. Moreover, this mechanical lysis of the membrane is an alternative and simple method for inner cell plasma extraction compared to electrokinetic, sonic and electroporative techniques (see Andersson and van den Berg (2003) for a review on 'cellomics').

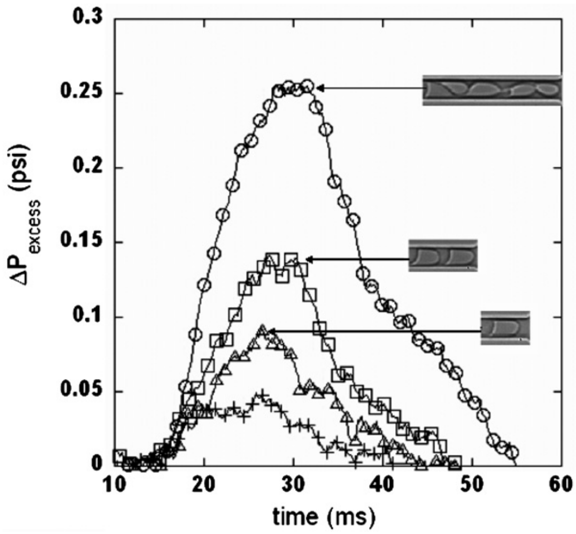

Figure 9. Pressure drop versus time for different conditions characterizing the state of the RBCs; the driving pressure is 5 psi. (+) healthy RBC; open symbols: RBCs treated with 0.001% glutaraldehyde; (△) one RBC; (□) a train of two RBCs; (○) a train of five RBCs.

4. Flows of suspensions of RBCs through a constriction

An improved understanding of flows in the microcirculation is crucial to obtain new insights into the origin of cardiovascular diseases, potential treatments, the distribution of cell types as a function of the various mechanical and chemical stresses, etc. Although much is known in the fluid dynamics and physics literatures about the flow of suspensions quite generally, much less is known when the suspended particles are deformable as is the case for RBCs. It has long been known that there are some unusual hydrodynamic characteristics of suspensions of blood cells, as Fahraeus was one of the first to describe (Goldsmith *et al* 1989). For example, the effective viscosity of the blood is observed to apparently be smaller for flow in smaller channels. We now understand that this effect arises because the cell suspension is not uniformly filling the entire channel, but rather, due to the drift of the deformable red cells away from the channel boundaries, the near-wall region is occupied by the low-viscosity plasma. We have visualized and quantified this drift for the flow of a suspension of cells in an abrupt constriction: hydrodynamically this configuration for the flow should model any kind of blockage fixed to the wall of a capillary, or may represent the obstruction caused in a small channel when a physician inserts a probe into a patient. For a study illustrating the drift away from a wall of single deformable vesicle in a shear flow, see Abkarian *et al* (2002).

4.1. Symmetrical geometries and symmetrical streamlines

We first illustrate a basic fact of viscous, low Reynolds number flows: when pressure-driven flow occurs past a symmetrical

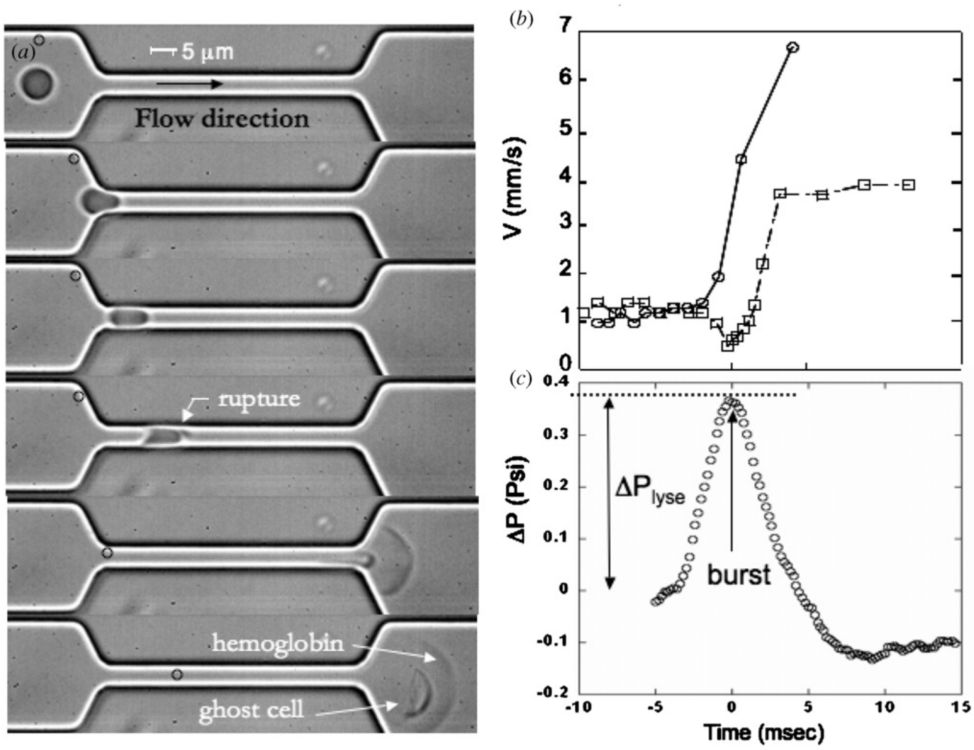

Figure 10. (*a*) Sequence of deformation and lysis of an inflated red cell in a 60 μm long and 5 μm wide channel. (*b*) Linear velocity of the centers of mass of an inflated RBC and in addition a small tracer particle (indicated in image (*a*) with a black circle). (*c*) Excess pressure drop associated with the deformation and bursting of the cell in the channel.

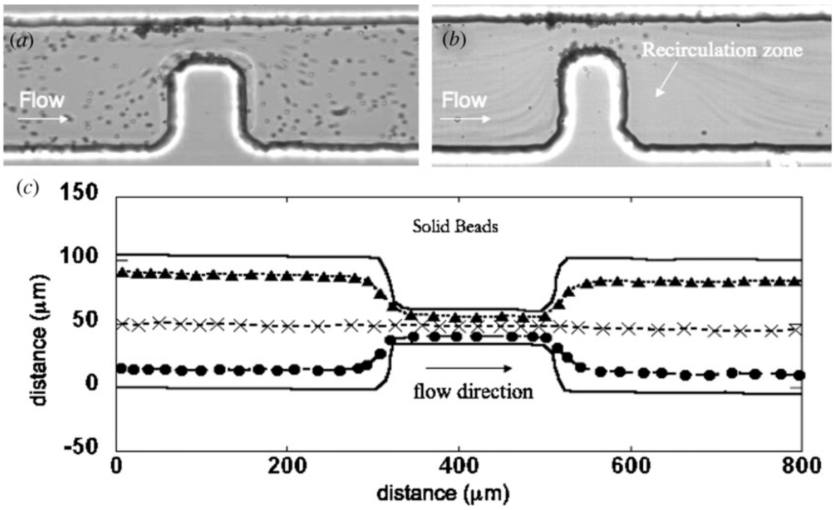

Figure 11. Low Reynolds number flow of polystyrene particles (1 μm diameter), over a rectangular step (size 25 μm × 75 μm) (a) at $Re = 0.001$, the streamline pattern is symmetrical left to right, (*b*) at $Re \sim 1$, the streamline pattern is asymmetrical left to right as a small eddy forms downstream of the obstruction. (*c*) Pressure-driven flow in a microchannel containing a symmetrical constriction (height 75 μm throughout). Suspended rigid spherical particles (8.7 μm in diameter) are tracked in the flow and the trajectories for the particles are shown. Each particle has a path that is essentially symmetrical about the center of the constriction (Faivre *et al* 2006).

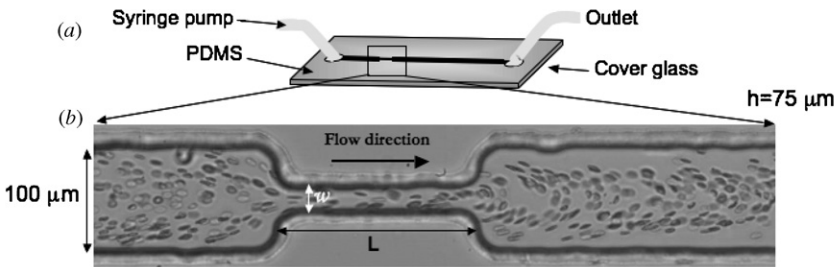

Figure 12. Microfluidic experiment for the flow of red blood cells through a constriction. The channel has a height of 75 μm and a width of 100 μm; the length of the constriction $L = 200$ μm, and its width $w = 25$ μm. (*a*) Schematic of the side view. (*b*) Top view image of the experiment.

object, the streamlines are fore-aft symmetric as shown in figure 11(*a*). Therefore, it is not possible from simple inspection of the streamlines to determine whether the flow is moving from left to right or from right to left. When the Reynolds number is higher, which is mostly easily achieved in the experiments by increasing the flow speed, the flow has a definite asymmetry with 'wake-like' features appearing downstream of any fixed object (figure 11(*b*)).

We investigated the flow of cells in microfluidic channel networks designed with a narrow constriction (figure 11(*c*)). First, we illustrate how rigid particles behave in these constricted configurations. As a consequence of the low Reynolds number, symmetrical streamline pattern, the trajectories of suspended rigid, neutrally buoyant spheres are also fore-aft symmetric (figure 11(*c*)). Thus, for such particles there is no tendency during or after flowing through the constriction for the drift of particles across streamlines.

4.2. Drift of deformable objects across streamlines

However, when we flow a suspension of RBCs through the constriction (figure 12) we observe a significant difference upstream as compared to downstream. In figure 12(*b*) we observe that upstream there is perhaps a region with low cell density that is comparable to the cell size adjacent to the boundary. This region of diminished concentration is what is commonly referred to as the 'cell-free layer'. Downstream of the constriction there is a much enhanced cell-free layer, which, for the concentrations used in our experiments, remains largely unchanged even a few centimeters downstream. This dramatic influence of the constriction illustrates some of the dynamical processes at play when the microstructure (in this case the cell) is deformable.

We systematically studied this cross-streamline drift as a function of the average shear rate in the constriction by varying the width of the constriction (Faivre *et al* 2006). Hydrodynamic considerations show that the drift speed is proportional to the (higher) shear rate in the constriction. Furthermore, we examined the influence of the duration of the shear rate by varying the length of the constriction; the cell-free layer monotonically increases in thickness for longer constrictions, essentially because there is additional time for cells to drift across streamlines. Nevertheless, it should be noted that there is only a weak effect of the flow rate on the development of the cell-free region: although a higher flow rate means that there is a higher shear rate in the constriction,

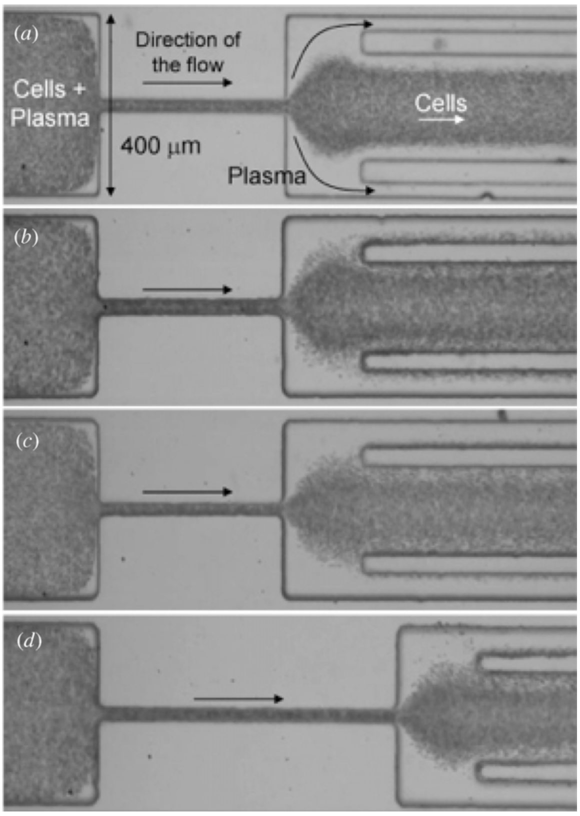

Figure 13. Microfluidic experiments with a constriction to separate blood cells from plasma. (a) $w = 25\ \mu$m, $L = 500\ \mu$m, $Q = 200\ \mu$l h^{-1}; outermost channel width = 30 μm. (*b*) Width of the outermost channel = 50 μm; otherwise the same conditions as (*a*). (*c*) The width of the outermost channel = 50 μm; otherwise the same conditions as (*a*) except that the constriction width = 15 μm. (*d*) The width of the outermost channel = 50 μm; constriction dimensions are $w = 25\ \mu$m, $L = 500\ \mu$m; $Q = 200\ \mu$l h^{-1} (Faivre *et al* 2006).

it also means that the cells spend less time on average in the constriction, and the two effects tend to cancel. These quantitative results were also compared with an elementary model for the drift speed of cells in shear flows (Faivre *et al* 2006).

4.3. Application: separation of the plasma

Because we have demonstrated that there is a systematic drift of cells toward the center of the channel whenever individual cells or a dilute suspension is flowed through a constriction, we recognized that it might be possible to harness this response to effect separation of the cells from the plasma. The cross-streamline drift is enhanced at higher shear rates, which should be expected for narrower channels, and acts for a time comparable to the length of the narrow channel. Such separations are illustrated in figure 13, where we show a device with three channels in parallel downstream of the constriction. In the central-most channel we collect almost all of the blood cells and plasma, while the two outer-most channels are almost if not completely free of RBC flows. As shown in figure 13, the separation is improved for narrower and/or longer constrictions. For example, for the experimental conditions shown in figure 13(*a*), we estimated that the blood cell concentration was increased about 24% in the central downstream channel.

5. Conclusions

In this paper we have summarized a number of different studies of cellular-scale flows in confined spaces. Our studies include detailed characterizations of the shapes of cells as a function of geometrical and flow parameters and the development of a manometer-like device for reporting changes of pressure when cells, or other small objects, obstruct the flow. Finally, a study at the scale of a suspension of blood cells examined the influence of a narrow constriction and quantified the significantly enhanced cell-free layer. We hope to have demonstrated how microfluidic tools can both bring new insights into the understanding of some aspects of physiological and pathological flows and provide new ways to explore mechanical properties of cells not accessible otherwise.

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Appendix: Experimental procedures

Blood Preparation. In our experiments the RBCs are extracted from a droplet of blood obtained by pricking the finger of a healthy donor. Prior to the experiment, an individual blood sample was diluted and washed twice with a solution of phosphate buffer saline (PBS) at a physiological osmolarity of 300 mOsmol. The solutions containing the blood cells were verified to be at pH 7.4 and were made with dextran of molecular weight of 2×10^6 at one of three concentrations, 4, 6 or 9% w/w, which correspond to viscosities of 11 cp, 20 cp and 47 cp respectively (water has a viscosity of 1 cp).

Experimental setup: glass capillaries and soft lithography. We pulled circular glass capillaries with different final radii using a standard micropipette puller. The final internal radii of the capillaries thus prepared were between 6 and 20 μm. An individual capillary is then connected to a syringe via the appropriate choice of tubing. The microfluidic devices were produced using common soft-lithography techniques (Duffy *et al* 1998.).

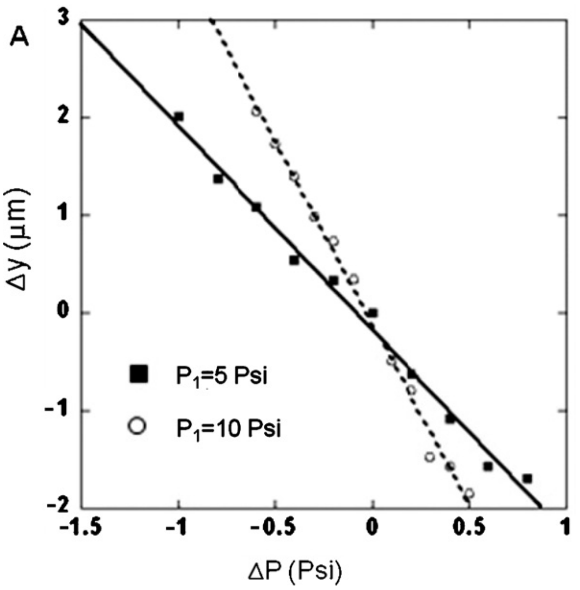

Figure 14. Variation of ΔY as a function of the change in pressure ΔP in the upper channel for two different upstream pressures P_1: (○) at 5 psi and (■) at 10 psi.

Fluid pumping. The fluid is set into motion by one of the two similar methods: either a specified flow rate is applied using syringe pumps or a specified pressure is applied by attaching the liquid container to a pressurized gas chamber. These two methods allow the application of a different range of speeds. When the system operates at steady state, these two distinct methods for moving the fluid are equivalent, but in the case of transitory flows, the controlled pressure method has a faster response than the flow-rate controlled method (Futterer *et al* 2004).

Manometer calibration. Before a campaign of measurements, each of the twin channels needs to be calibrated in pressure. The flow is produced by pressurizing fluid-filled syringes connected to the two inlets of the device. In the absence of RBCs, the pressure P_1 applied in the upper test channel and the pressure P_2 in the lower control channel are fixed so that the fluid–fluid interface downstream is centered in the main exit channel (figure 7(*a*)). We change the pressure P_1 in small increments ΔP without changing the pressure P_2 in the control channel and follow the displacement of the interface in the Y direction by performing image analysis with Matlab software (figure 7(*b*)). The variation ΔY is linear in ΔP for the two initial working pressures applied, $P_1 = 5$ psi and $P_1 = 10$ psi as presented in figure 14. The slope $\Delta Y(\Delta P)$, which refers to as calibration slope at $P_1 = 5$ psi is twice as large as the slope at $P_1 = 10$ psi (in absolute value) and the two curves lie on each other by renormalizing ΔY by a, which is half the width of the exit channel, and ΔP by P_1 as shown in figure 14; both responses are expected for small variations of this pressure-driven viscous flow.

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